Additive Manufacturing and Metallization of Feeds for Radio Astronomy Antennas

Nicolás Seguel[1]

[1] *Pontificia Universidad Catolica de Chile*

July 30, 2026

Abstract

Abstract text goes here.

Keywords: additive manufacturing, metallization, electroplating, antennas, radio astronomy

Contents

1 Introduction **2**

2 Additive Manufacturing **2**

3 Metallization of Plastic Parts **2**

4 Electroplating **2**

5 Direct Metal Manufacturing (SLM) **2**

6 Characterization Methodology **2**

7 Results **2**

8 Conclusions **2**

1 Introduction

Introduction text goes here.

Revisar la introducción y agregar referencias relevantes.

2 Additive Manufacturing

Text on additive manufacturing methods explored (SLA, SLM, FDM).

3 Metallization of Plastic Parts

Text on metallic paints and their role as a conductive base.

4 Electroplating

Text on the electroplating process using copper and silver baths.

5 Direct Metal Manufacturing (SLM)

Text on 316L stainless steel printing.

6 Characterization Methodology

Text on S11, S12, VSWR, Smith Chart, radiation pattern, and mechanical tests.

7 Results

Preliminary results go here.

8 Conclusions

Conclusions and future work go here.